Kepler had noticed a haunting regularity in the heavens: the farther a planet orbits from the Sun, the longer its year — and the relationship is exact. Squaring the periodic time T always tracks the cube of the greater axis a, so that $T^2 \propto a^3$. Equivalently, the period grows as the $3/2$ power — what Newton calls the sesquiplicate ratio — of the axis. What Kepler found only by patient observation, Newton now proves from the inverse-square law: "the periodic times in ellipses are in the sesquiplicate ratio of their greater axes."